

Hope Artificial Intelligence

Scenario Based Learning

A company works with number of employees; all the works are dependents on the employees. Even if one of the employees resign the job immediately then assigned work will be not finished at the time, so delivery of the project to the clients will be delayed. Company planned to make solution for this, they want to know which employee may resign next. If they know previously, they can arrange alternative to avoid such problem. As an AI Engineer you must give Solution to this.

- A) How will you achieve this in AI?
- B) Find out the 3 -Stage of Problem Identification
- C) Name the project
- D) Create the dummy Dataset.

Dummy dataset :

S.No.	Emp. Name	Emp. Code	DOJ	Designation	Grade	Role	Project Name	Last appraisal rating 1 - 5	Exp.in years
1	Sundar M	100451	21-Aug-20	VP	7	Tower Head	Head of projects	4	5.7
2	Ramesh M	102519	17-Apr-21	AVP	6	Project Head	Head of projects	2	5.0
3	Vimalkumar S	104348	5-Jun-21	Project Manager	4	FullStack	FinLea	3	4.9
4	Pragadheesh S	106128	11-Mar-22	Project Manager	4	Architect	SwisTek	2	4.1
5	Magesh T	107528	10-Sep-22	Project Manager	4	SWE	AvivaStocks	4	3.6
6	Deepak V	107629	20-Apr-22	Jr.Consultant	1	SWE	AvivaStocks	2	4.0
7	Kamesh V	109090	5-May-22	Sr.Consultant	3	SWE	AvivaStocks	4	4.0
8	Vijayakumar A	110536	2-Nov-22	Jr.Consultant	1	FullStack	FinLea	4	3.5
9	Chandrika K	111801	13-Feb-23	Sr.Consultant	2	FullStack	FinLea	2	3.2
10	Aadhikesavan H	112963	1-Jan-24	Jr.Consultant	1	FullStack	FinLea	2	2.3
11	Dhanasekar B	113803	15-Jan-24	Sr.Consultant	2	Architect	SwisTek	2	2.3
12	Bharathkumar J	114629	19-Feb-25	Jr.Consultant	1	Architect	SwisTek	3	1.2
13	Priyanka V	115054	24-May-25	Sr.Consultant	2	SWE	MainStream	0	0.9

Stage 1: Identify Domain:

Here they given numerical output, so we choose the Machine Learning,

Stage 2: Check input & output:

We have both input and output hence we choose the Supervised Learning.

Stage 2: Choose suitable model:

The output data is in numerical values hence we choose the Regression.

Logic:

Problem statement, check the last appraisal rating of employees and tenure they are serving in the company.

If the employee served less than 2 years and the rating is less than 3 in last appraisal.
and

If the employee served more than 2 years and the rating is less than 3 in last appraisal.

The above two if statements are fulfilled then we can easily narrow down the next resignation may expect in which project.

In addition, we roll out the survey about their last appraisal rating to know their EQ

AI Solution:

If statements will give us the clear data about the rating and tenure

And the EQ score if low in survey, then we could see the satisfaction level of employees.

We may expect them to drop their resignation.

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If the employee served more than 2 years and the rating is less than 3 in last appraisal.

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6	Deepak V	107629	20-Apr-22	Jr.Consultant	1	SWE	AvivaStocks	2	4.0
4	Pragadheesh S	106128	11-Mar-22	Project Manager	4	Architect	SwisTek	2	4.1